

Mathematical Proof of Equivalence between CAM and Grad-CAM in ResNet Architecture and Layer-wise Visualization Pattern Analysis across Learning Rates

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Abstract

This study aims to verify the reliability of CAM and Grad-CAM as analytical tools by proving their mathematical and experimental equivalence, and to analyze the characteristics of Weakly Supervised Object Localization (WSOL) through visualization pattern changes across internal layers and learning rates. Using a ResNet-50 model fine-tuned on the Stanford Dogs dataset, we confirmed that the weights of the two techniques are exactly proportional by a constant factor ($\frac{1}{2}$) at the final layer (Layer 4), yielding identical visualization results and localization performance (IoU: 0.5347) after normalization. Based on this established reliability, we applied Grad-CAM to Layers 1–4 and observed feature maps across learning rates (0.1, 0.01, 0.001). The results revealed that as layers deepen or learning rates are optimized, activations progressively concentrate from the ‘entire object’ to ‘class-discriminative regions (e.g., dog face).’ We conclude that dynamic threshold adjustment and multi-layer information aggregation are essential for more accurate object localization.

1 Introduction

1.1 Background: Structural Limitations of CAM and the Versatility of Grad-CAM

Despite their high predictive performance, deep learning models suffer from the ‘black-box’ problem where internal decision-making processes remain opaque, driving active research in Explainable Artificial Intelligence (XAI). Among XAI methods, CAM (Class Activation Map) [2] provides intuitive visualization results but is structurally constrained to the final layer where GAP (Global Average Pooling) [1] is applied. In contrast, Grad-CAM [3] leverages gradient information to be universally applicable to all internal layers (Layers 1–4) of a model.

1.2 Problem Statement

Existing studies visualize model decision rationale through CAM and Grad-CAM, yet quantitative analysis of errors and limitations arising from binarizing heatmaps to extract Bounding Boxes (BBox) remains relatively insufficient. In particular, in-depth diagnosis of how activation patterns in internal layers change under varying hyperparameter conditions (e.g., learning rate) is required.

1.3 Research Objectives: Equivalence Proof and Internal Layer Analysis

To trust Grad-CAM as an internal model analysis tool, one must first prove that the two techniques produce mathematically and experimentally identical results at the final layer (GAP), where both can be commonly applied. Once Grad-CAM’s reliability is established at the final layer, it provides logical justification for qualitatively analyzing ‘layer-wise activation patterns’ and ‘changes in model decision criteria under hyperparameter variations’—analyses that were impossible with CAM alone. This study seeks to explore avenues for improving localization performance through such multi-faceted analysis.

2 Related Work & Background

2.1 GAP and CAM

GAP (Global Average Pooling), proposed by Lin et al. (2013) [1], is a key technique that replaces FC layers to preserve spatial information in the Network In Network architecture. Given a spatial size of $X \times Y$ with total pixel count Z ($Z = X \times Y$), the result F_k of applying GAP to the k -th feature map $f_k(x, y)$ of the last convolutional layer is defined as:

$$F_k = \frac{1}{Z} \sum_{x,y} f_k(x, y) \quad (1)$$

Zhou et al. (2016) [2] proposed CAM, which uses F_k to derive an activation map for a specific class c . The class score S_c for class c is given by:

$$S_c = \sum_k w_k^c F_k = \sum_k w_k^c \left(\frac{1}{Z} \sum_{x,y} f_k(x, y) \right) = \frac{1}{Z} \sum_{x,y} \sum_k w_k^c f_k(x, y) \quad (2)$$

Therefore, the degree of activation (CAM) for class c at a specific spatial location (x, y) is derived as $\frac{1}{Z} \sum_k w_k^c f_k(x, y)$.

2.2 Grad-CAM and Mathematical Equivalence

Grad-CAM, proposed by Selvaraju et al. (2017) [3], utilizes backpropagation gradients to overcome the structural limitation of CAM that requires GAP. The importance weight α_k^c representing the influence of the k -th feature map A^k on the score y^c for a specific class c is defined as:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{i,j}^k} \quad (3)$$

In architectures such as ResNet-50 [5], where a GAP operation and linear classifier follow immediately after the last convolutional layer A^k , the class score y^c is $y^c = \sum_k w_k^c \left(\frac{1}{Z} \sum_{i,j} A_{i,j}^k \right)$. Taking the partial derivative with respect to feature map pixel $A_{i,j}^k$ yields:

$$\frac{\partial y^c}{\partial A_{i,j}^k} = \frac{w_k^c}{Z} \quad (4)$$

Consequently, computing the Grad-CAM weight α_k^c :

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{w_k^c}{Z} = \frac{1}{Z} \left(Z \cdot \frac{w_k^c}{Z} \right) = \frac{w_k^c}{Z} \quad (5)$$

Thus, the Grad-CAM weight α_k^c is not exactly equal to the CAM weight w_k^c , but is precisely proportional by a constant factor of $\frac{1}{Z}$. Since the activation map (heatmap) undergoes Min-Max Normalization for visualization, this constant scaling difference is canceled out, and the final visualization results and extracted BBoxes become perfectly identical. This mathematically proves that the two techniques produce essentially equivalent results at the final layer.

3 Experimental Setup

3.1 Model Configuration and Dataset

This study is based on the ResNet-50 [5] architecture pre-trained on ImageNet. The final FC layer was modified to accommodate 120 classes for object recognition, and fine-tuning was performed using the Stanford Dogs dataset [4]. Training employed the SGD optimizer (Momentum=0.9) with a batch size of 32 for a total of 10 epochs. Three learning rate conditions were set for comparison: 0.1, 0.01, and 0.001. During data preprocessing, images were resized to 224×224 and normalized using ImageNet statistics.

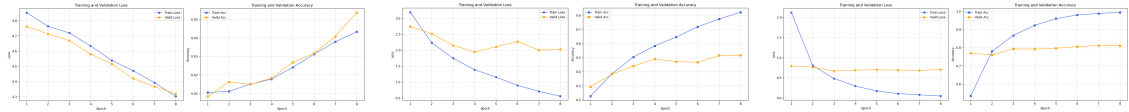
3.2 Evaluation Metric

The generated activation maps (heatmaps) were resized to the original image dimensions, and rectangular BBoxes were generated based on pixel coordinates exceeding the threshold ($\theta = 0.3$). The threshold of 0.3 was determined through preliminary heuristic experiments as the optimal value for the most stable extraction of object contours. Localization performance was quantitatively evaluated using IoU (Intersection over Union) between the predicted BBox and Ground Truth BBox.

4 Results & Analysis

4.1 Learning Dynamics across Learning Rates

Training dynamics were observed under various learning rate (LR) conditions. The smallest learning rate (0.001) achieved the most stable and highest performance.



(a) LR : 0.1, Train - Loss : 4.3020, Acc : 4.32%, Val - Loss : 4.3188, Acc : 5.36%
 (b) LR : 0.01, Train - Loss : 0.5559, Acc : 82.20%, Val - Loss : 2.0265, Acc : 51.62%
 (c) LR : 0.001, Train - Loss : 0.0487, Acc : 99.38%, Val - Loss : 0.7079, Acc : 81.11%

Figure 1: Learning Dynamics across Learning Rates

4.2 Mathematical Equivalence of Visualization Techniques

Comparing the CAM and Grad-CAM visualization heatmaps extracted from the final layer (Layer 4) of ResNet-50, the Mean Absolute Error (MAE) between the two heatmaps after Min-Max Normalization was confirmed to be 0, indicating perfect agreement. The BBox IoU evaluation also yielded identical performance of 0.5347 for both techniques,

demonstrating that the mathematical proportionality proven in Section 2.2 converges to equivalent results in practical experimental settings after normalization.

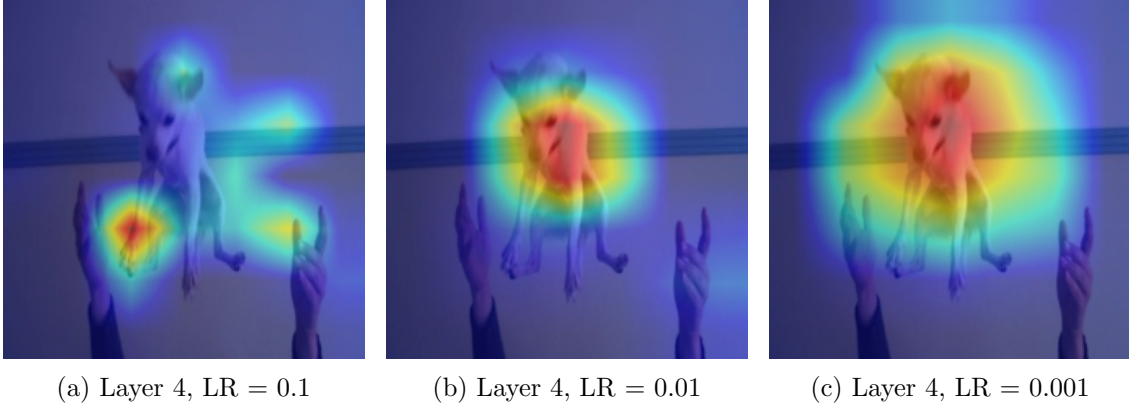


Figure 2: Mathematical Equivalence of Visualization Techniques (Layer 4)

4.3 Layer-wise Localization Performance Analysis

Using the validated Grad-CAM, we analyzed activation patterns across internal layers (Layers 1–4) and learning rate variations. A clear trend of activation concentration on specific regions was observed as layer depth increases.

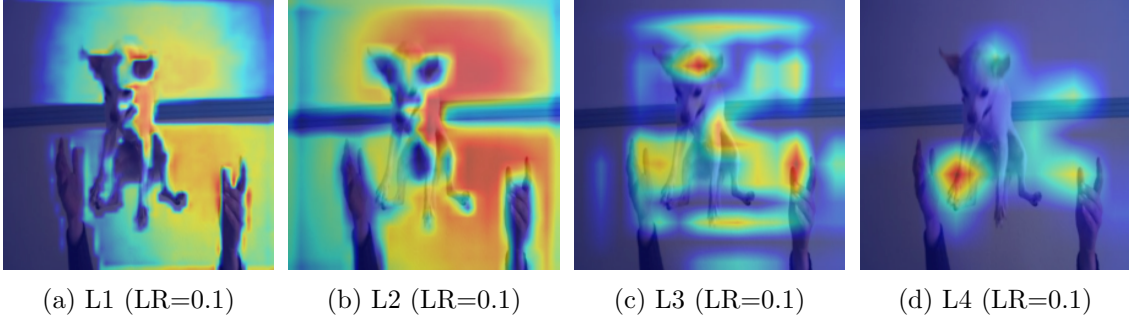


Figure 3: Layer-wise Localization Performance Analysis (LR=0.1)

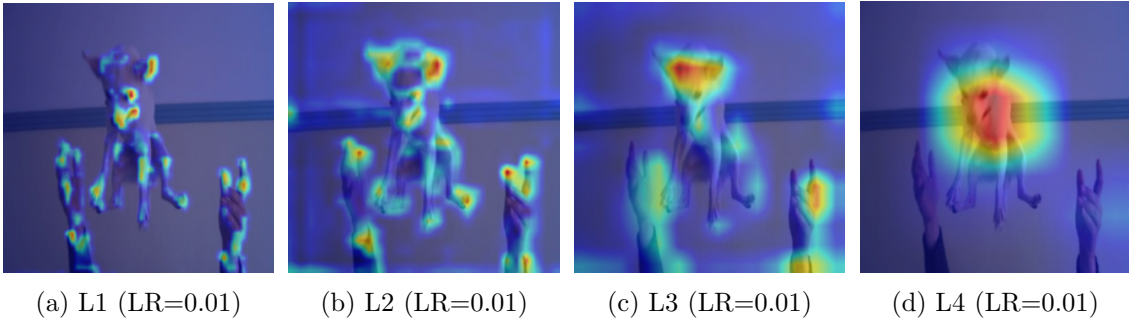


Figure 4: Layer-wise Localization Performance Analysis (LR=0.01)

- **Layers 1 & 2 (Shallow layers):** Respond to low-level features such as edges and textures, exhibiting widespread noisy activations across broad regions.

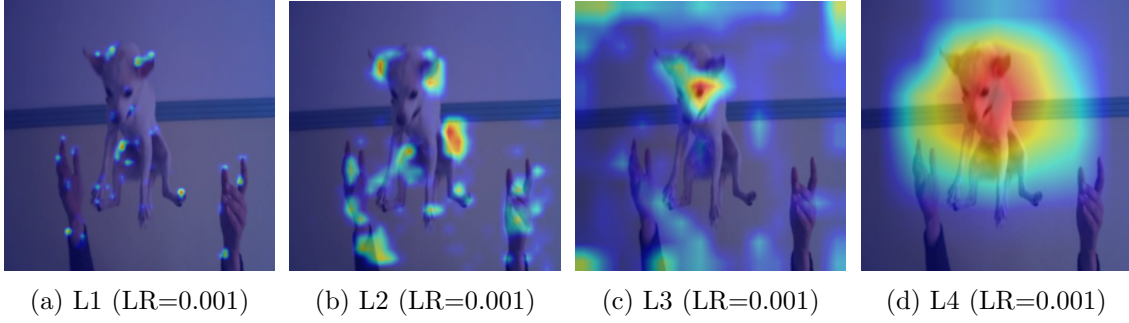


Figure 5: Layer-wise Localization Performance Analysis (LR=0.001)

- **Layer 3 (Intermediate layer):** Captures the overall shape of the object to a reasonable extent and contains substantial spatial information favorable for BBox extraction.
- **Layer 4 (Deep layer):** Rather than the entire object, activations concentrate strongly on the most discriminative parts for classification (e.g., dog face).

5 Discussion & Conclusion

5.1 Interpretation of Results

Synthesizing the experimental results, the fundamental reason for the limited localization accuracy (IoU) lies not in algorithmic flaws but in the **inherent limitation of classifiers**. Classification-oriented CNNs do not need to capture the entire object boundary; they minimize loss by identifying only the most discriminative features (e.g., face) and terminate learning accordingly. This implies that achieving accurate localization solely from classification labels is an inherently challenging task in the WSOL (Weakly Supervised Object Localization) domain, and raises the need for auxiliary localization signals separate from classification loss.

5.2 Limitations

Using a fixed threshold poses risks in localization. Setting the threshold too low introduces noise, while setting it too high causes excessive BBox shrinkage. Since this experiment relied on a single dataset (Stanford Dogs), additional validation on other domains is required.

5.3 Conclusion

This study proved the mathematical equivalence of CAM and Grad-CAM in the ResNet-50 model and conducted an in-depth analysis of activation patterns across learning rates and layers. Proper learning rate selection and understanding of layer-wise features serve as essential foundations for enhancing deep learning model transparency and improving object localization performance. For future work, we propose an ensemble approach that aggregates the holistic shape information from Layer 3 with the core semantic information from Layer 4.

References

- 1 Lin, M., Chen, Q., & Yan, S. (2013). Network in network. *arXiv preprint arXiv:1312.4400*.
- 2 Zhou, B., Khosla, A., Lapedriza, A., Oliva, A., & Torralba, A. (2016). Learning deep features for discriminative localization. *CVPR*.
- 3 Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-cam: Visual explanations from deep networks via gradient-based localization. *ICCV*.
- 4 Khosla, A., Jayadevaprakash, N., Yao, B., & Fei-Fei, L. (2011). Novel dataset for fine-grained image categorization: Stanford dogs. *CVPR Workshop*.
- 5 He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *CVPR*.